

XYZ CoWork Inc.

Data Mining Project Guidelines

Fall Semester 2026-2027

Last Updated: 02 of September 2026

1 Introduction

Consumers today are becoming more selective about the business they support and where they spend their money. This makes it vital for companies to develop a deep understanding of their customer base in order to tailor their services and marketing strategies more effectively.

XYZ CoWork Inc. (XYZ) faces exactly this challenge with its diverse member base. In a competitive coworking industry, understanding member segments is key to improving satisfaction, retention, and revenue. This process, known as customer segmentation [3, 1], allows XYZ to divide customers into smaller groups that share similar characteristics or behaviors. For instance, some members prioritize premium services and dedicated space, while others are cost-conscious and focused on basic desk access. By uncovering these patterns, XYZ can tailor its services and membership perks to meet the unique needs and behaviors of each segment.

2 Project Overview

In this project, you will act as consultants for XYZ. Your task is to analyze coworking membership data and corresponding space-usage activity collected over a three-year period [2]. Using these insights, you will develop a data-driven segmentation strategy. The dataset description is provided in Section 8.2.

The segmentation should be approached from multiple perspectives, such as:

- *Value-based segmentation*, grouping customers according to their economic contribution.
- *Behavioral segmentation*, analyzing usage habits and space-attendance behaviors.
- *Other perspectives that are found relevant.*

The ultimate objective is to integrate these perspectives into a final segmentation framework that supports XYZ in crafting a comprehensive marketing strategy.

To structure the project, we recommend applying the **CRISP-DM** methodology [4, 5], progressing systematically from business understanding and data preparation through modeling and evaluation.

By the end of the project, students are expected to gain proficiency in:

- Applying unsupervised learning techniques for customer segmentation
- Conducting multi-perspective segmentation analysis
- Translating technical insights into actionable business strategies

The project will be carried out in groups of up to four students. It will consist of three sequential deliverables, moving from exploratory data analysis to the development of strategic recommendations.

2.1 Deliverables Summary

Deliverable Components	Points	Description
Deliverable 1: EDA	20	4 November 2026
Jupyter Notebook	5	GroupXX_EDA_Code.ipynb
Discussion Slides	15	GroupXX_EDA_Slides.pdf
Deliverable 2: Clustering	60	3 January 2027
Jupyter Notebook	5	GroupXX_Clustering_Code.ipynb
Report	55	GroupXX_Clustering_Report.pdf
Deliverable 3: Discussion	20	Date TBA (January 2027)
Individual Assessment	20	In Person

3 References

- [1] Sara Dolnicar, Bettina Grün, and Friedrich Leisch. *Market Segmentation Analysis*. Jan. 2018. DOI: [10.1007/978-981-10-8818-6](https://doi.org/10.1007/978-981-10-8818-6). URL: <https://doi.org/10.1007/978-981-10-8818-6>.
- [2] *Guide to IBM Cognos Analytics Sample Data*. Dec. 2024. URL: <https://community.ibm.com/community/user/browse/blogs/blogviewer?BlogKey=FF811D76-ABE0-4DF2-BCEA-917176FD72E4>.
- [3] Christiana Jolaoso. *Customer segmentation: the ultimate guide*. June 2024. URL: <https://www.forbes.com/advisor/business/customer-segmentation/>.
- [4] Christoph Schröer, Felix Kruse, and Jorge Marx Gómez. "A Systematic Literature Review on Applying CRISP-DM Process Model". In: *Procedia Computer Science* 181 (2021). CENTERIS 2020 - International Conference on ENTERprise Information Systems / ProjMAN 2020 - International Conference on Project MANagement / HCist 2020 - International Conference on Health and Social Care Information Systems and Technologies 2020, CENTERIS/ProjMAN/HCist 2020, pp. 526–534. ISSN: 1877-0509. DOI: [10.1016/j.procs.2021.01.199](https://www.sciencedirect.com/science/article/pii/S1877050921002416). URL: <https://www.sciencedirect.com/science/article/pii/S1877050921002416>.
- [5] Rüdiger Wirth and Jochen Hipp. "CRISP-DM: Towards a standard process model for data mining". In: *Proceedings of the 4th international conference on the practical applications of knowledge discovery and data mining*. Vol. 1. Manchester. 2000, pp. 29–39. URL: <http://www.cs.unibo.it/~daniilo.montesi/CBD/Beatriz/10.1.1.198.5133.pdf>.

4 Deliverable 1: Exploratory Data Analysis (20 points)

This deliverable lays the groundwork for segmentation by examining the coworking membership dataset in detail. The focus is on uncovering meaningful patterns, identifying limitations, and generating initial hypotheses about customer groups.

Key Tasks:

- Conduct descriptive statistics and visualizations to highlight distributions, trends, and anomalies, while noting which variables appear most relevant for segmentation.
- Assess data quality issues and evaluate how these may affect clustering reliability.
- Identify preliminary behavioral signals that suggest distinct types of customers.
- Develop and justify engineered features. Show how these derived variables capture richer aspects of customer behavior and explain their potential contribution to clustering models.

In presenting results, consider addressing the following:

1. Which findings were most unexpected or insightful, and what do they reveal about likely customer clusters?
2. What data limitations pose the greatest risks for clustering, and how might they be mitigated?
3. Which patterns in customer activity, including those revealed by engineered features, suggest natural groupings, and what cluster characteristics do you anticipate?
4. How would you communicate these insights to non-technical stakeholders? Include a clear explanation of the expected number of clusters, the most important differentiating features, and any anticipated challenges.

4.1 Component 1: Jupyter Notebook (5 points)

This component assesses technical execution skills through documented code and analysis workflow.

Expectations:

- Jupyter notebook following file naming format `GroupXX_EDA_Code.ipynb`
- All code cells executed with visible outputs
- Systematic data exploration workflow
- Organized code, using Markdown headings to separate sections
- Clean, well-documented code with consistent variable naming
- Error-free execution in fresh environment

4.2 Component 2: Discussion Slides (15 points)

This component evaluates analytical thinking and business insight as you transform technical exploration into strategic insights, following CRISP-DM methodology to prepare for segmentation analysis.

Expectations:

- Report following file naming format `GroupXX_EDA_Slides.pdf`
- Minimum 3, maximum 5 slides of content, **excluding the cover slide**
- Slides should showcase the most relevant findings: insights and interpretations, not just data descriptions
- Submitted before the discussion session, alongside the code

After submitting the discussion slides, each group will be scheduled for a 10-15 minute discussion with the instructors. During this discussion, students will answer questions about their analysis and explain their methodologies.

Target schedule: 09-13 November 2026. This is **not yet final** and will be confirmed later on Moodle.

5 Deliverable 2: Clustering Analysis (60 points)

This phase applies clustering techniques to the coworking membership dataset to generate meaningful customer segments. The emphasis is on experimenting with multiple perspectives, validating results, and merging insights into a comprehensive solution that supports business objectives.

Key Tasks:

- Prepare the dataset for modeling.
- Analyze available features to determine which segmentation perspectives can be meaningfully applied.
- Perform segmentation using the identified perspectives.
- Apply at least two clustering approaches within each perspective and compare results.
- Propose a final merged segmentation solution that integrates the most important insights across perspectives into a coherent framework.

When presenting results, consider addressing the following:

1. Which clustering method(s) produced the most interpretable and stable results for each perspective?
2. How many clusters best represent the customer base overall, and what evidence supports this decision?
3. What differentiating features emerged as most important within each perspective, and how do they complement one another in the merged solution?
4. How would you describe the final set of customer segments to a business audience? Summarize defining traits, potential marketing opportunities, and any challenges encountered when integrating perspectives.

5.1 Component 1: Jupyter Notebook (5 points)

This component demonstrates technical implementation of clustering methodology through documented code workflow.

Expectations:

- Jupyter notebook following file naming format GroupXX_Clustering_Code.ipynb
- All code cells executed with visible outputs
- Clear code organization using Markdown structure
- Multiple clustering algorithms implemented and compared
- Clean, reproducible code with consistent documentation
- Error-free execution

5.2 Component 2: Report (55 points)

This component evaluates analytical rigor and business insight as you integrate technical clustering results with strategic recommendations for XYZ's customer segmentation.

Expectations:

- Report following file naming format GroupXX_Clustering_Report.pdf
- **Maximum** of 10 pages of content using provided framework and template
- Insights and interpretations (not just bulleted list of what you did)
- Evidence-based segment characterization and strategic recommendations
- Required annexes: AI usage attribution + contribution statement + responsibility statement

Report Framework

Executive Summary: Provide concise overview of key segment findings, strategic recommendations for each customer group. Target executive audience requiring actionable insights.

Introduction: Establish context by connecting Phase 1 findings to clustering objectives. Document data preparation decisions and feature engineering based on Phase 1 findings.

Methodology: Present feature selection process, algorithm selection process, parameter tuning methodology, and implementation approach comparing multiple clustering methods.

Results & Validation: Analyze clustering performance through technical metrics and statistical significance testing. Interpret business meaning of discovered segments with customer profiles and behavioral characteristics.

Strategic Recommendations: Develop actionable strategies for each customer segment, including marketing approaches, service customization, and loyalty program optimization for XYZ.

Conclusion: Integrate technical findings with business strategy, address methodology limitations, and outline future research directions for XYZ's segmentation approach.

Annexes: The following are mandatory and do not count towards page limit. Examples are provided at the end of this document (Section [8.1](#)).

- AI Usage Statement
- Contribution Statement
- Responsibility Statement

6 General Policies

6.1 Group composition

- Maximum of FOUR (4) students in each group. We recommend a group size of three (3).
- ALL students must be enrolled in a group on Moodle, regardless of group size.
- Students must be enrolled in a group on Moodle before the first delivery deadline.
- Changes to the group composition after the first delivery date is not recommended.

6.2 Plagiarism check

All submitted reports will undergo a plagiarism check. Ensure that your work is original and properly cite any external sources used.

6.3 Use of Generative AI tools

- The use of generative AI tools is permitted but must be fully disclosed. It is essential that the students' own contributions to the report exceed that of any AI tools used.
- Students must include a section in the annex of the report documenting the use of AI tools (Annex AI Usage Statement)
- Students must document specific AI tool usage (ChatGPT, Claude, Gemini, etc.)
- Students must specify how AI tool was used (ideation, code assistance, refining, proofreading)
- If the group did not use AI tools, this statement must explicitly state this.
- Students are fully responsible for the contents of the report they submit, including any material generated or assisted by AI tools.
- Do not just copy and paste the results if AI tools are used.
- Students may be asked to explain the meaning of any AI-generated content submitted to ensure their comprehension.

All insights, business interpretations, and strategic recommendations must represent original group analysis.

Penalties applied

A 10% penalty will be applied for each day of delay in delivering the project. Further deductions to the deliverable grade may also be applied for deviations to the guidelines.

7 Optional Bonus Components

Each deliverable can include an optional bonus component up to 20% of that deliverable's points. Bonus components are designed to reward exceptional work while maintaining core assignment focus.

7.1 Deliverable 1 Options

*Select up to two options, each contributing **up to** 10% additional credit toward the deliverable grade.*

1. External Data Integration & Validation

- Source and integrate relevant external dataset (industry benchmarks, demographic census data, etc.)
- Validate findings against external sources
- Provide comparative context for customer behavior patterns
- Include data source credibility assessment

2. Multidimensional Outlier & Anomaly Detection

- Apply advanced anomaly detection methods.
- Identify unusual customer behaviors that may distort clustering.
- Discuss whether anomalies should be excluded, downweighted, or analyzed separately.

3. Interactive EDA Dashboard

- Build a lightweight interactive dashboard (e.g., Plotly Dash, Streamlit) for exploring customer attributes.
- Include basic filtering, drill-down, and visualization of distributions.
- Provide stakeholders with a tool for self-exploration of the dataset.

7.2 Deliverable 2 Options

*Select up to two options, each contributing **up to** 10% additional credit toward the deliverable grade.*

Discussion of advanced business-focused analysis (2 pages maximum + Jupyter notebook)

1. Financial Impact Modeling (2 pages maximum + supporting code)

- Calculate ROI projections for each segment
- Include customer lifetime value estimates
- Provide cost-benefit analysis of implementation
- Create investment timeline with expected returns

Implementation and discussion of advanced clustering techniques (2 pages maximum + Jupyter notebook)

2. Fuzzy Clustering Implementation

- Apply fuzzy c-means with membership degree analysis
- Compare hard vs soft clustering interpretations
- Analyze customer overlap between segments
- Include uncertainty quantification in business recommendations

3. Deep Embedded Clustering (Autoencoders)

- Implement autoencoder-based dimensionality reduction
- Apply clustering in learned latent space
- Compare with traditional feature-based clustering
- Visualize learned representations and cluster boundaries

Implementation and demonstration of interactive visualization (max 5-minute video + code hosted on GitHub)

4. Interactive Cluster Visualization Dashboard

- 3D cluster visualization with rotation and zoom capabilities
- Real-time filtering by perspectives
- Customer detail pop-ups with segment characteristics
- Export functionality for stakeholder sharing

5. Dynamic Parameter Tuning Interface

- Live cluster parameter adjustment (K-values, distance metrics)
- Real-time visualization of cluster changes
- Performance metric updates during parameter modification
- Business impact calculator for different configurations

6. Interactive Dendrogram for Multi-view Clustering

- Hierarchical clustering across multiple data perspectives
- Clickable tree navigation with cluster detail expansion
- Perspective switching (demographic vs behavioral vs temporal)
- Cross-perspective cluster comparison tools

7.3 General Bonus Guidelines

7.3.1 Quality Standards

- Bonus work should maintain the same high quality as your main deliverables.
- Completing bonus tasks should never compromise your core assignment requirements.
- Aim to add meaningful value that goes beyond the minimum expectations.

7.3.2 Documentation

- Clearly document all bonus components so others can understand your work.
- Explain why you chose this particular enhancement and what it contributes.
- Share any challenges you faced and how you solved them.

7.3.3 Grading

- Bonus points are calculated as a percentage of your base deliverable score.
- Exceptional work that meets all criteria earns full bonus points.
- Partial credit can be awarded for well-executed but incomplete bonus work.
- **Bonus points are only awarded if the core deliverable meets at least the passing standard.**

8 Annex

8.1 Annex Statements Examples

Annex AI Usage Statement

Outlines specific AI tool usage (ChatGPT, Claude, Copilot, etc.) and specify how each tool was used (ideation, code assistance, refining, proofreading, etc.)

AI tools were used for coding syntax assistance (ChatGPT for pandas operations), literature review brainstorming (Claude for industry context ideas), and report proofreading (Grammarly for grammar checks). All analytical insights, business interpretations, and strategic recommendations represent original group analysis and thinking.

Annex Contribution Statement

Individual student contributions to deliverables.

Sample Student (Student ID: 20267777)

- Fitness industry research for Business Understanding
- Introduction section, poster design
- Logistic Regression implementation

Lorem Student (Student ID: 20268888)

- Feature engineering ideation, RFM calculations
- Gym visit frequency analysis, peak hours identification
- Abstract, Conclusion sections, report coordination

Student Ipsum (Student ID: 20269999)

- Data undersampling approach
- Culinary preference insights
- Results interpretation, notebook organization

All members contributed to collaborative discussions and ideation.

Annex Responsibility Statement

We, the group members listed above, certify that this report represents our original analytical work and interpretations. While AI tools were used as specified above, all insights, conclusions, and recommendations are the result of our independent analysis and critical thinking. We take full responsibility for the accuracy and quality of this submission.

8.2 Annex: Dataset Information

Variable	Description
member_id	Unique identifier of each member
age	Age of the member at the extraction date.
gender	Gender of the member
country	The country where the member (or their associated company) is based.
industry	The industry sector the member works in.
is_team_account	Whether the membership is held by a team/company account rather than an individual
n_colleagues_on_account	Number of colleagues sharing the account.
plan_tier	Current membership tier. (hot_desk < flex < dedicated_desk < full_time < private_office)
plan_days_per_week	Number of days per week the member's plan entitles them to use the space.
membership_start	Date the membership first began.
membership_end	Date the membership first ended.
last_badge_swipe	Date of the member's most recent physical entry to the space.
days_since_last_visit	Number of days since the member's last recorded visit.
total_visits	Total number of visits the member made over the whole membership.
avg_weekly_visits	Average number of visits per week over the membership.
entitled_visits_last_period	Number of visits the member's plan entitled them to during the last period of their membership.
actual_visits_last_period	Number of visits the member actually made during the last period of their membership.
meeting_room_hours	Hours of meeting-room usage recorded for the member.
event_space_hours	Hours of event-space usage recorded for the member.
guest_passes_issued	Number of guest passes the member issued.
uses_meeting_rooms	Whether the member ever used the meeting-room add-on.
uses_event_space	Whether the member ever used the event-space add-on.
uses_guest_passes	Whether the member ever used the guest-pass add-on.
uses_print	Whether the member used the print add-on.
uses_mail_handling	Whether the member used the mail-handling add-on.
uses_phone_booths	Whether the member used the phone-booth add-on.

uses_day_passes	Whether the member used the day-pass add-on.
uses_storage	Whether the member used the storage add-on.
uses_bike_locker	Whether the member used the bike-locker add-on .
n_renewals	Number of times the membership was renewed.
lifetime_value	Total revenue attributed to the member over the whole membership.
